# Evaluation runs: chain-replication (chain-replication)

generated 2026-11-03 from brain-data b2ae60f86 2026-11-03 17:57:09 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 6 | 18 | conforms | 0.1 | 2 | 4 h, 250 GB | 6 | 21 | brain01, 2026-11-02 |
| 3,3 | 90 | 1260 | conforms | 0.7 | 4 | 4 h, 250 GB | 100 | 4507 | brain01, 2026-11-02 |

## stateful

### stateful: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 3,2 | 6 | 18 | conforms | 0.0 | 4 | 4 h, 250 GB | brain01, 2026-11-03 |
| 3,3 | 90 | 1260 | conforms | 0.2 | 4 | 4 h, 250 GB | brain01, 2026-11-03 |

## offline

### offline: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 6 | 18 | conforms | 0.1 | 4 | 1 h, 250 GB | 0.0 | 0.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3,3 | 90 | 1260 | conforms | 0.2 | 4 | 1 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 6 | 18 | conforms | 5.0 | 164 | 4 h, 250 GB | 0.0 | 5.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3,3 | 90 | 1260 | conforms | 9.3 | 367 | 4 h, 250 GB | 0.0 | 8.9 | 0.3 | merge | brain01, 2026-11-03 |

## cross online, Java implementation (JMC) against the Rust specification export

### cross online, Java implementation (JMC) against the Rust specification export: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 3,2 | 6 | 18 | conforms | 6.4 | 154 | 4 h, 250 GB | | brain01, 2026-11-03 |
| 3,3 | 90 | 1260 | conforms | 12.5 | 2504 | 4 h, 250 GB | | brain01, 2026-11-03 |

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 18 | 18 | conforms | 5.1 | 164 | 4 h, 250 GB | 5.0 | 0.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3,3 | 1260 | 1260 | conforms | 9.1 | 367 | 4 h, 250 GB | 8.9 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 18 | 18 | conforms | 5.1 | 164 | 4 h, 250 GB | 0.0 | 5.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3,3 | 1260 | 1260 | conforms | 9.4 | 367 | 4 h, 250 GB | 0.1 | 8.9 | 0.3 | merge | brain01, 2026-11-03 |

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 3,2 | 18 | 18 | conforms | 6.6 | 170 | 4 h, 250 GB | | brain01, 2026-11-03 |
| 3,3 | 1260 | 1260 | conforms | 13.0 | 2510 | 4 h, 250 GB | | brain01, 2026-11-03 |

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 3,2 | 18 | 18 | conforms | 0.0 | 4 | 4 h, 250 GB | | brain01, 2026-11-03 |
| 3,3 | 1260 | 1260 | conforms | 0.3 | 4 | 4 h, 250 GB | | brain01, 2026-11-03 |